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CHEM F111 : General Chemistry
Semester II: AY 2023-24

Lecture- 11, 7th February 2024, Wednesday

Last class

Determine the energy levels/terms/electronic states for a configuration [GS/ES] and arrange them in increasing order of energy.

Simple examples in course: [s^1], [p^1], [d^1], [s^1 p^1], [p^1 p^1], [p^1d^1], etc..

Without determining all the levels/states, one can directly determine the ground state term symbol for any configuration.

Explanation of spectra based on selection rules and term symbols

Recap: Term symbol: two p electrons (in different shells)

- A **term** is a combination of the one of the allowed total orbital angular momentum values (L_i) with one of the allowed total spin angular momentum values (S_i);

Ex:

One of the excited state configuration of C is [He]$2s^2 2p^1 3p^1$

Allowed total orbital angular momentum values

$l_1 = 1,\ l_2 = 1$

$$L = l_1 + l_2,\ l_1 + l_2 - 1,\ l_1 + l_2 - 2, \ldots, |l_1 - l_2|.$$

$$L = (1+1=2),\ (1+1-1=1),\ |1-1=0|.$$

$$L = 2(D),\ 1\ (P),\ 0\ (S)$$

DIFFERENT POSSIBLE RELATIVE ORIENTATIONS OF THE RESULTANT ORBITAL ANGULAR MOMENTA OF A 2-ELECTRON SYSTEM

Recap: Ground state term symbol determination

d^3/d^7 Configuration:

three electrons should be in parallel spin

For max. S $s_1 = ½$

$s_2 = ½$

$s_3 = ½$

Max. S = 3/2

=2S+1 = 4

3 (F) = Max. L possible

$M_L = 2 + 1 + 0$

	m_l				
	+2	+1	0	-1	+2
m_s					

Ground state term: ^4F

For a ^4F term, J can be calculated

L = 3 and S = 3/2,

So, J = 9/2,7/2,5/2,3/2.

$^4F_{9/2}$

$^4F_{7/2}$

$^4F_{5/2}$

$^4F_{3/2}$

Ground state energy level (d^7 conf.)

Ground state energy level (d^3 conf.)

Question: The Sodium D emission line spectra is actually a doublet, *i.e.*, two very closely spaced lines. Explain.

Answer: This means transition from excited conf. of Na ($3p^1$) to ground level conf. of Sodium ($3s^1$)

Due to spin-orbit coupling the $3p^1$ (l = 1, s = ½) configuration actually splits into two levels, corresponding to total angular momentum quantum number *j*= *J* = 3/2, and 1/2 : $^2P_{3/2}$ & $^2P_{1/2}$

$3s^1$ (l = 0, s = ½) configuration corresponds to total angular momentum quantum number *j* =J = 1/2 (l + s): $^2S_{1/2}$

Two lines are due to transition: $^2P_{3/2} \rightarrow {}^2S_{1/2}$ $\Delta L = 1, \Delta S = 0, \Delta J = 1$

$^2P_{1/2} \rightarrow {}^2S_{1/2}$ $\Delta L = 1, \Delta S = 0, \Delta J = 0$

Chemical Bond

Description of electronic structure of molecules

Ionic bond: *Transfer of electron(s) from one atom to another, and the consequent interaction (electrostatic) between the ions so formed.*

Covalent bond: *Two neighboring atoms share a pair of electrons. (G. N. Lewis)*

Major approaches to the calculation of molecular structure:

VSEPR model (Valence Shell Electron Pair Repulsion) **(self-study)**

Theories

1. Valence Bond Theory (NOT IN SYLLABUS)

2. Molecular Orbital Theory

Both the theories use wavefunctions (orbitals)

Chemical Bond

Valence bond theory:

- **Extension of Lewis electron dot model**
- **Overlap of atomic orbitals and sharing of electron pairs**
- **Works fine for many systems**
- **Limited to two center two electron bonds**

  ***Delocalization*: Resonance**
- **Cannot describe excited states**

Molecular orbital theory:

- **Electron(s) moving in the joint field of nuclei**
- **Set up the Hamiltonian: Exactly solvable for H_2^+, but not for more complex molecules**
- **Molecular orbitals: Linear combination of atomic orbitals (LCAO)**
- **Can handle delocalization, excited states: A general theory**

Chemical bond theories: Molecular Hamiltonian

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$$\hat{H} = -\frac{\hbar^2}{2}\sum_{A=1}^{M}\frac{1}{m_A}\nabla_A^2$$ Nuclear kinetic energy (T_n)

$$-\frac{\hbar^2}{2m_e}\sum_{i=1}^{N}\nabla_i^2$$ Electronic kinetic energy (T_e)

$$+\sum_{A=1}^{M-1}\sum_{B=A+1}^{M}\frac{Z_A Z_B e^2}{4\pi\varepsilon_0|\vec{r}_A-\vec{r}_B|}$$ Inter-Nuclear repulsion energy (V_{nn})

$$-\sum_{A=1}^{M}\sum_{i=1}^{N}\frac{Z_A e^2}{4\pi\varepsilon_0|\vec{r}_i-\vec{r}_A|}$$ Electron-nuclear attraction energy (V_{ne})

$$+\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\frac{e^2}{4\pi\varepsilon_0|\vec{r}_i-\vec{r}_j|}$$ Inter-Electronic repulsion energy (V_{ee})

Schrödinger equation using this form of Hamiltonian is intractable

Born Oppenheimer (BO) approximation

Nuclei are much more massive than electrons. Electronic motions are on much faster time scales as compared to the nuclear motions. Electrons can adjust immediately to instantaneous changes in the positions of nuclei.

The BOA says that the total molecular wavefunction can be accurately expressed as nuclear wavefunction multiplied by electronic wavefunction (almost multiplicative separability).

Solve the electronic Schrödinger equation for a series of fixed positions of the nuclei.

$$\hat{H}_e(\{\vec{r}_n\})\,\psi_e(\{\vec{r}_e\},\{\vec{r}_n\})=E_e(\{\vec{r}_n\})\,\psi_e(\{\vec{r}_e\},\{\vec{r}_n\})$$

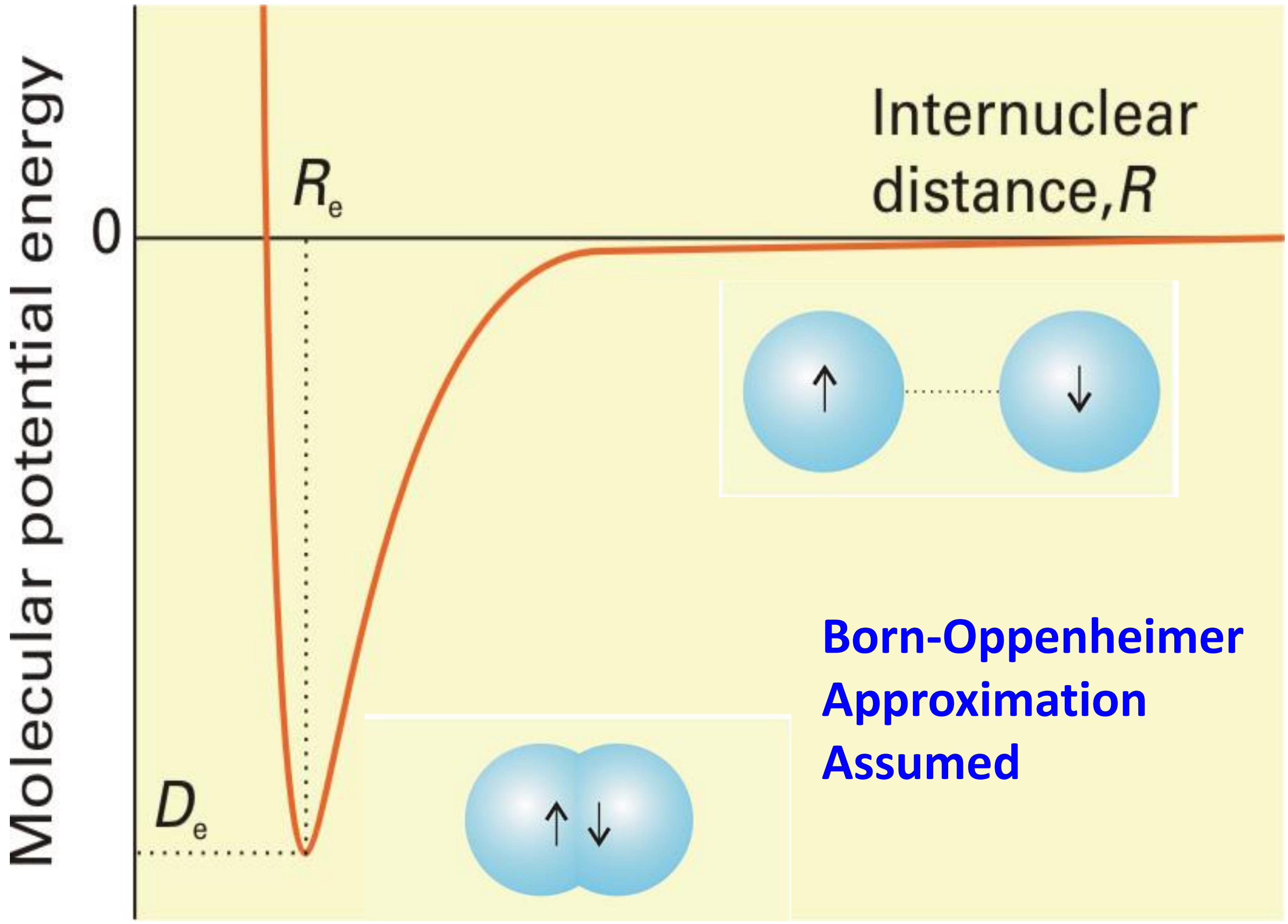
$$\hat{H}_e=\hat{T}_e+\hat{V}_{ne}+\hat{V}_{ee}+[\hat{V}_{nn}]$$

Born Oppenheimer (BO) approximation

The electronic energy or molecular potential energy $E_e(\{\bar{r}_{\underline{n}}\})$ is obtained for various nuclear positions for each electronic state and appears as the potential energy for the nuclear Hamiltonian. The nuclear Schrödinger equation gives energy states corresponding to the translations, rotations and vibrations in the molecule.

$$[\hat{T}_n + E_e(\{\vec{r}_n\})]\,\psi_n(\{\vec{r}_n\}) = E_n\,\psi_n(\{\vec{r}_n\})$$

Potential energy curve of a diatomic molecule



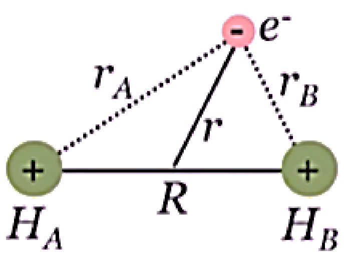
Born-Oppenheimer Approximation Assumed

Molecular orbital theory [for diatomic molecules]

We follow approach similar to what we did for atoms.

For example, consider the H_2^+ molecule – it is the simplest single-electron diatomic molecule.

The Hamiltonian operator for $\mathbf{H_2^+}$



$$\widehat{H}\left(H_2^+\right) = -\frac{\hbar^2}{2m_A}\nabla_A^2 - \frac{\hbar^2}{2m_B}\nabla_B^2 - \frac{\hbar^2}{2m_e}\nabla_e^2 - Q\frac{e^2}{r_A} - Q\frac{e^2}{r_B} + Q\frac{e^2}{R}$$

- The Hamiltonian operator for $\mathbf{H_2^+}$ in Born Oppenheimer (B.O.) approximation:

$$\widehat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r_A} - \frac{e^2}{4\pi\varepsilon_0 r_B} + \frac{e^2}{4\pi\varepsilon_0 R}$$

where r_A and r_B are the distances of the electron from nucleus A and B, respectively, and R is the internuclear separation, which we treat as a variable parameter.

Molecular orbital theory [for diatomic molecules]

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- The Hamiltonian operator for $\mathbf{H_2^+}$ in Born Oppenheimer (B.O.) approximation:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r_A} - \frac{e^2}{4\pi\varepsilon_0 r_B} + \frac{e^2}{4\pi\varepsilon_0 R}$$

In principle, it is possible to obtain exact solutions to the Schrödinger equation of a one-electron diatomic molecule.

Being one-electron system, the Hamiltonian eigenfunctions, are one-electron functions of the molecule, are referred to as molecular orbitals (MO)s.

Molecular orbital theory [for diatomic molecules]

How about **many-electron** diatomic molecules?

Hamiltonian for a 2e molecule

$$\hat{H}_e = -\frac{\hbar^2}{2m_e}(\nabla_1^2 + \nabla_2^2) - \frac{Z_A e^2}{4\pi\varepsilon_0 r_{1A}} - \frac{Z_B e^2}{4\pi\varepsilon_0 r_{1B}} - \frac{Z_A e^2}{4\pi\varepsilon_0 r_{2A}} - \frac{Z_B e^2}{4\pi\varepsilon_0 r_{2B}} + \frac{e^2}{4\pi\varepsilon_0 r_{12}} + \frac{Z_A Z_B e^2}{4\pi\varepsilon_0 r_{AB}}$$

One can use orbital approximation and express the total wavefunction in terms of the anti-symmetrized product of the molecular orbitals.

$$\psi(1,2,\ldots) = \phi_i(1)\phi_j(2)\ldots \times spin\ part$$

Where, the ϕ_k's are the MOs. The spatial and spin parts are chosen carefully to ensure anti-symmetry in the many-electron wavefunction.

What about polyatomic molecules? – Need to obtain MOs by solving Schrödinger equation for one-electron system of each polyatomic protoype – highly impractical approach

Molecular orbital theory [for diatomic molecules]

We note that although exact solution to a single-electron molecule is possible, the exactness is of no use for many-electron atoms. Unlike atoms (where, all have spherical symmetry and hydrogenic atom can serve as excellent proto-type to calculate orbitals), due to variety of shapes in molecules, proto-type of one particular geometry would not work for all molecules.

Moreover, even if exact MO's of a given proto-type are obtained, the many-electron wavefunction will always be approximate. Thus, exact orbitals are not as such required to generate the many-electron molecular wavefunction.

Thus, it is useful to develop an approximate method for describing the MOs.

The chemical bonding is based on the concept of atoms coming close to each other and sharing the electrons to increase stability of the system.

Thus, it is a good idea (and, indeed, a good approximation) to express the MOs in terms of linear combination atomic orbitals (LCAO).

Molecular Orbital Theory

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- Assume that MOs are linear combination of atomic orbitals (LCAO-MO): $\Psi = \Sigma_i (C_i\varphi_i)$

- *n* Atomic Orbitals(φ) combined should give *n* MOs (Ψ)

- In a homonuclear diatomic species such as H_2^+

$\Psi = C_A\varphi_A + C_B\varphi_B$ (The electron is either in A or B)

Since the two hydrogen atoms and the corresponding 1s orbitals are identical, it is easy to both would contribute equally to the total MO probability density. This results in two possiblities in the coefficients, thereby resulting in two MOs (and the the wavefunctions):

Since electron can be found with equal probability in A or B,
$C_A^2 = C_B^2$ Or $C_B = \pm C_A$

- Possible wavefunctions for simple case H_2^+ as: $\Psi = N(\varphi_A \pm \varphi_B)$
- N is normalizing factor.

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Let's Consider first: $\Psi_1 = N(\varphi_A + \varphi_B)$

Born interpretation: **Probability density is proportional to the square of modulus of wave function.**

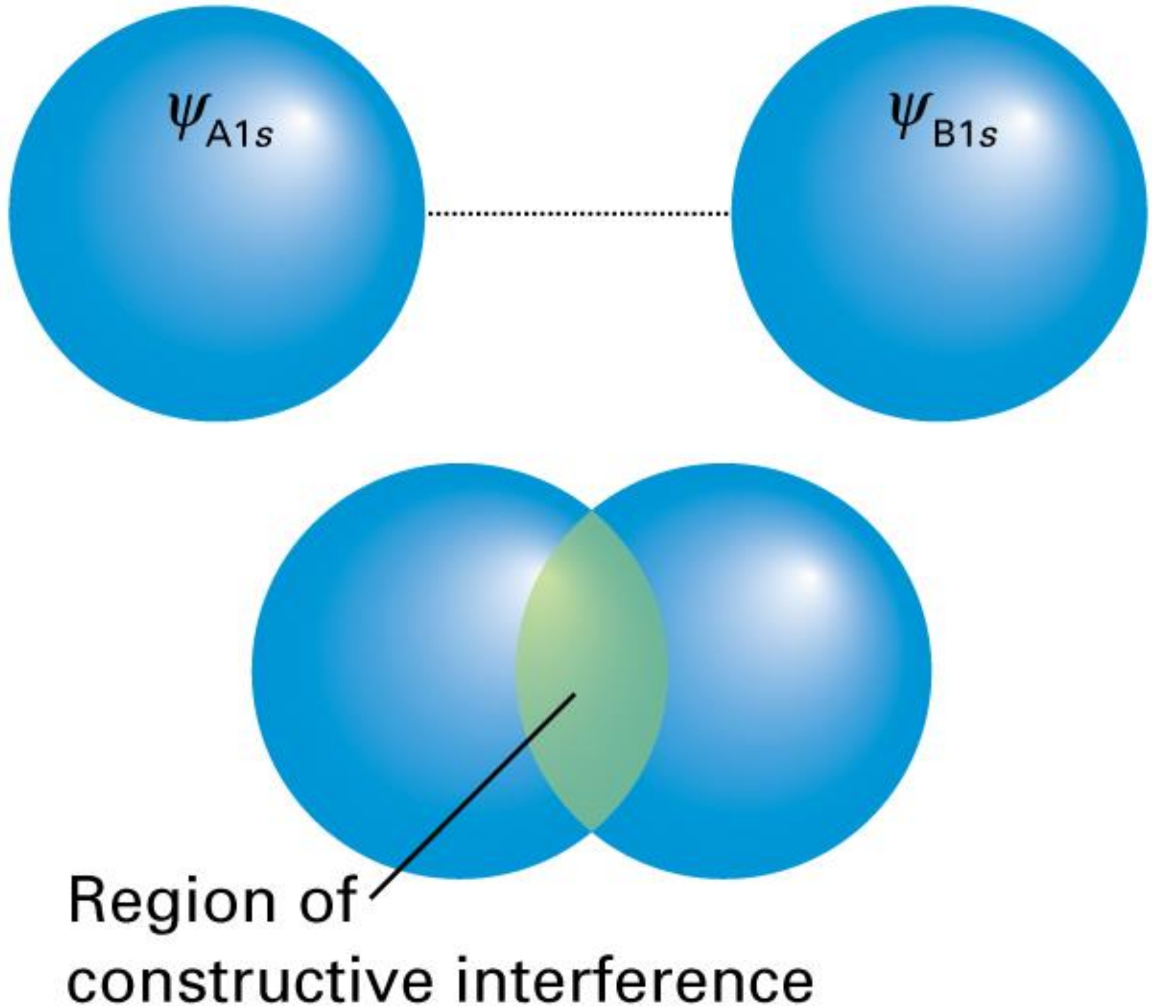
$$\Psi_1^2 = N^2 (\varphi_A^2 + \varphi_B^2 + 2\,\varphi_A \varphi_B)$$

Bonding MO ***(Energy is lowered for the electron if it is placed in this orbital)***

We write the configuration as $1\sigma^1$

1σ because this is lower most energy σ.

Molecular Orbital Theory

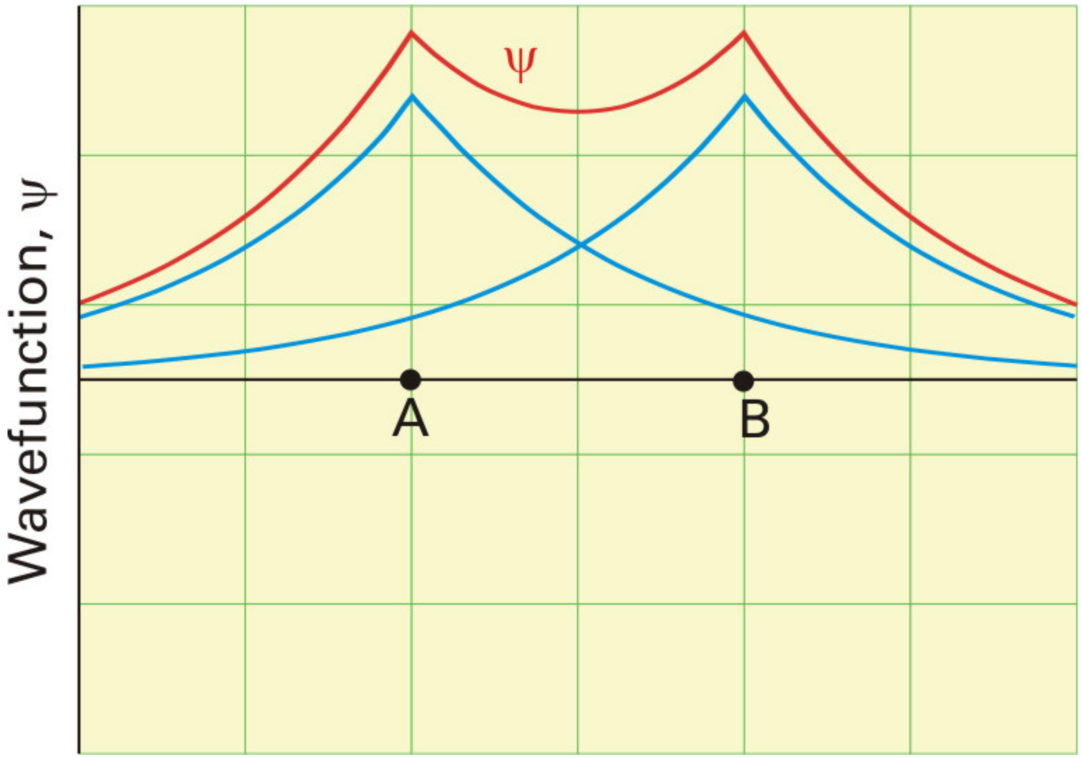


Bonding

Cylindrical symmetry around inter nuclear axis.

Molecular Orbital Theory

Variation of bonding MO ψ_1 along inter-nuclear axis



Bonding

$$\psi_1{}^2 = N^2 (\varphi_A{}^2 + 2\varphi_A\varphi_B + \varphi_B{}^2)$$

- **Lower energy; σ orbital –** resemblance with s orbital

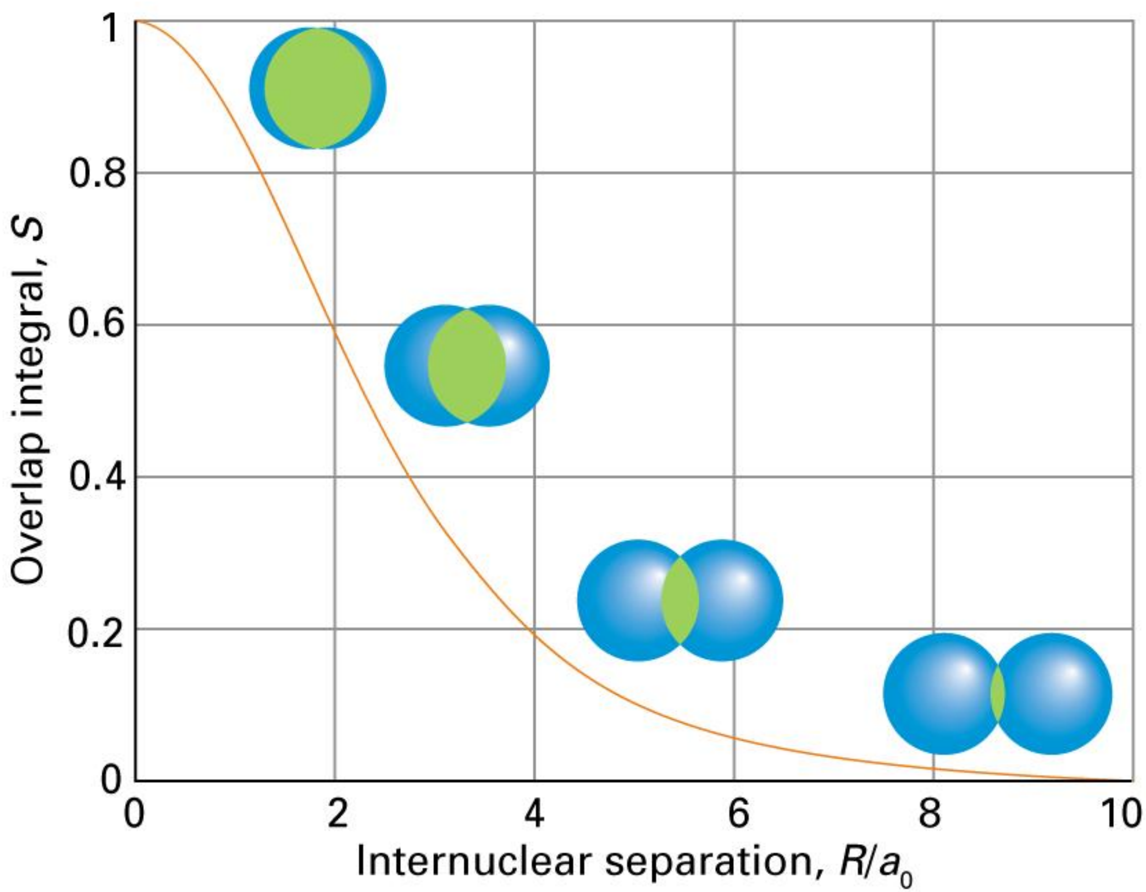
Molecular Orbital Theory

Overlap Integral :

$$S_{AB} = \int \psi_A \psi_B d\tau$$

S =1 perfect overlap
and S= 0 no overlap

[This will be discussed again in a latter slide]



$$S_{HH} = \{1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\} e^{-(R/a_0)}$$

Molecular Orbital Theory

Let's now Consider : $\Psi_1 = N(\varphi_A - \varphi_B)$

Born interpretation:

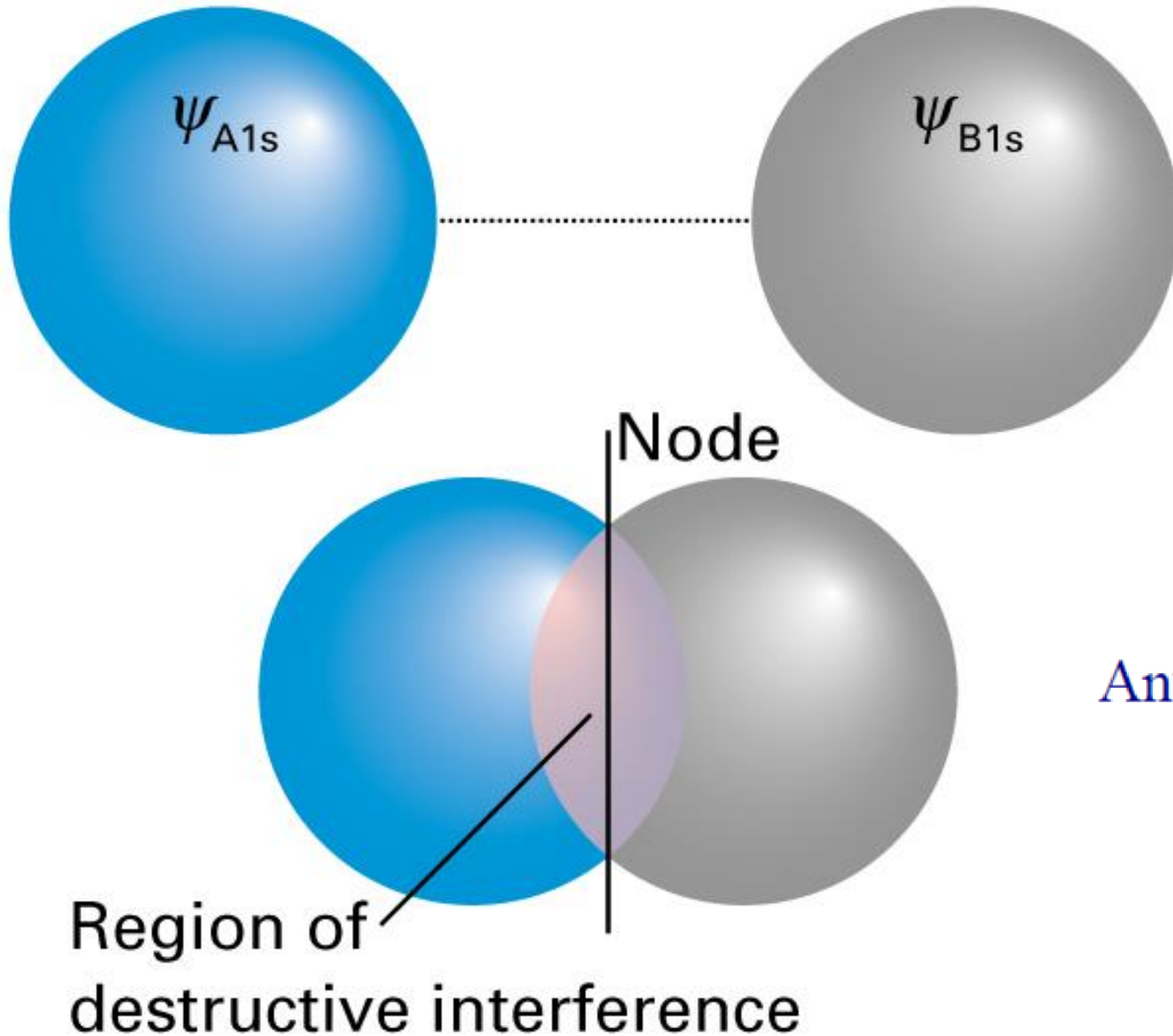
$$\Psi_1^2 = N^2 (\varphi_A^2 + \varphi_B^2 - 2\,\varphi_A \varphi_B)$$

Anti Bonding MO ***(Energy is raised for the electron if it is placed in this orbital)***

We write the configuration as $1\sigma^*$

σ^* **because this is excited state.**

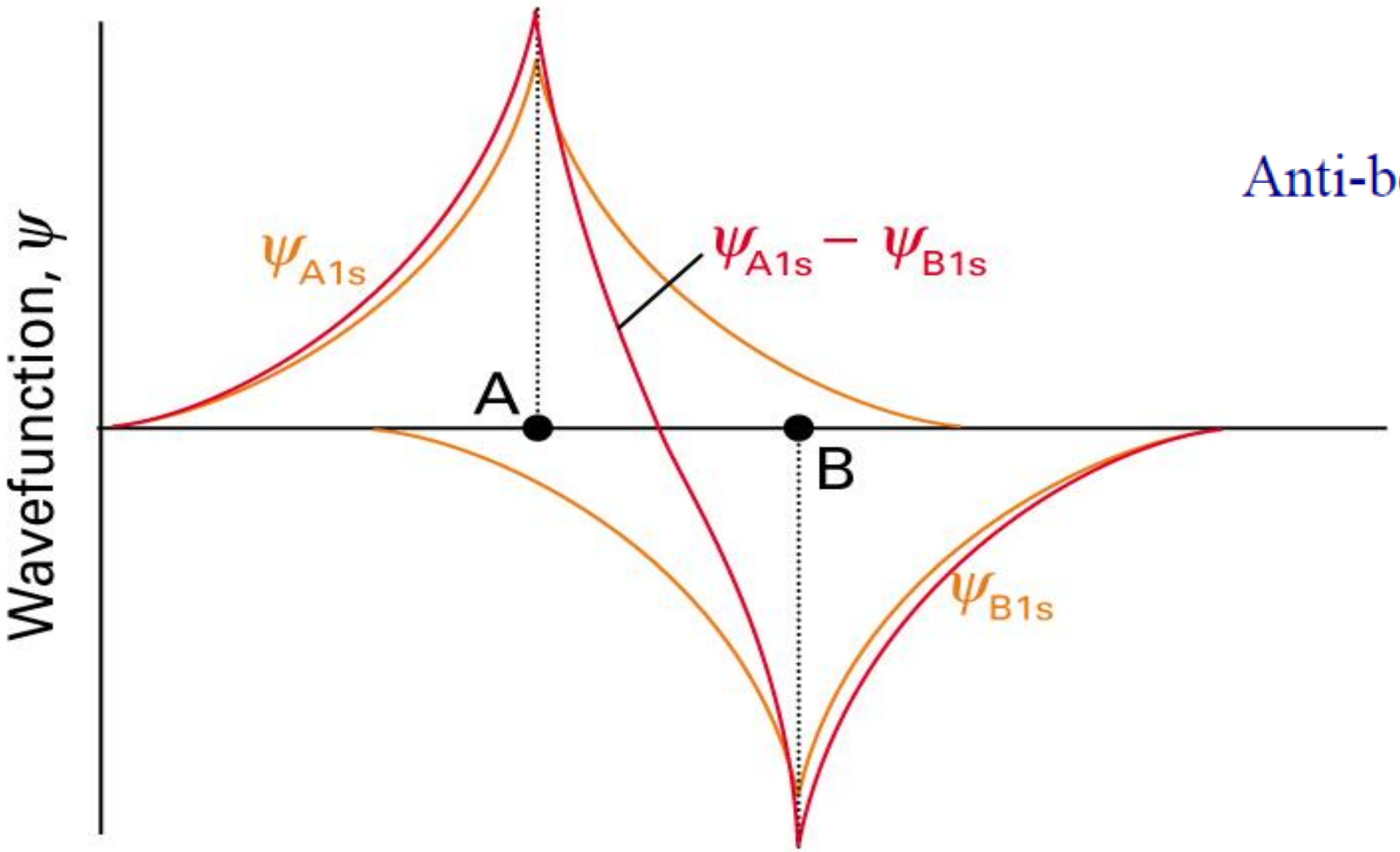
Molecular Orbital Theory



Anti-bonding

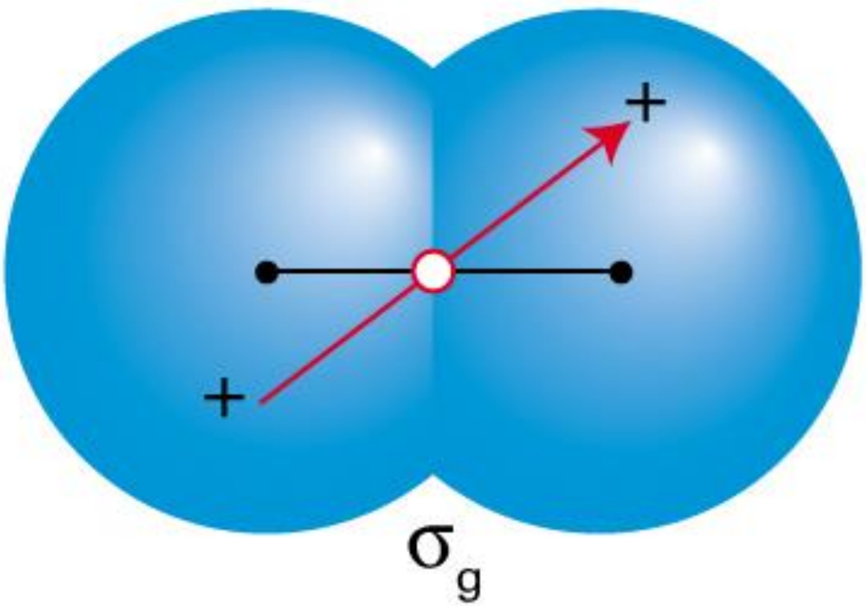
Nodal Plane Exists

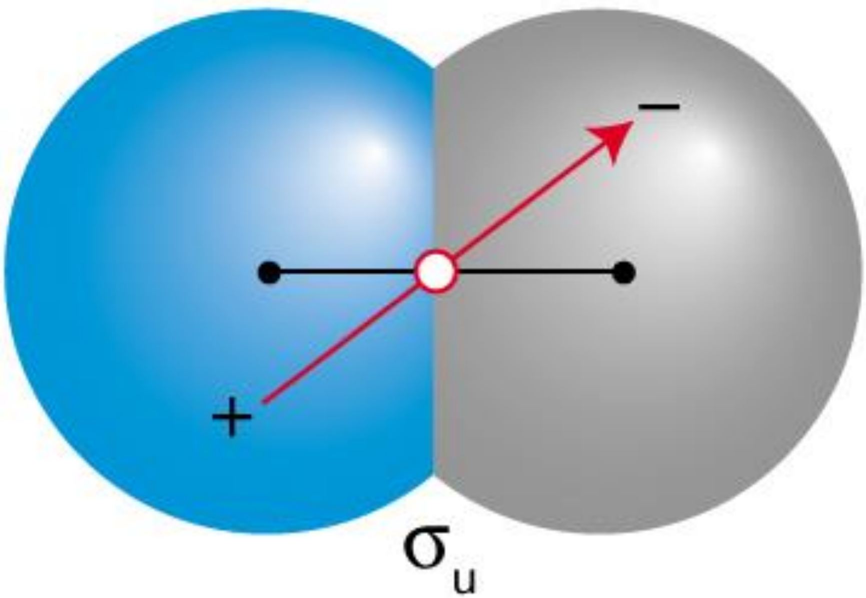
Molecular Orbital Theory



Anti-bonding

Molecular Orbital Theory





Bonding
(Even – g)

Anti-bonding
(odd – u)